Robust FLOWER Under Noise Model Mismatch

A Review and Controlled Reconstruction Study

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Abstract

FLOWER uses a pretrained flow-matching prior to reconstruct an image from a degraded measurement. Its measurement-refinement step depends explicitly on the assumed noise variance, making noise calibration an important practical question. We review the method and its mathematical arguments and investigate reconstruction quality when the assumed Gaussian noise standard deviation differs from the true value. The study uses 20 fixed images from each of CelebA and AFHQ-Cat, five inverse problems, and five assumed-to-true noise ratios, totaling 1,000 reconstructions. All ratios for a fixed image and task share their measurement and source-noise draws. Denoising is the most sensitive task; random inpainting is nearly insensitive over the tested range. Overestimation reduces mean PSNR in all ten dataset–task combinations, whereas underestimation has task-dependent effects, including improvements in super-resolution. PSNR, SSIM, and LPIPS reveal both agreement and perceptual trade-offs. These results show that robustness depends on the forward operator, actual noise level, and metric.

1 Introduction

An inverse problem aims to recover an unknown clean image from incomplete or noisy measurements. The acquisition process may discard pixels, reduce spatial resolution, or blur image structure. A useful reconstruction must balance consistency with the observation and knowledge of plausible natural images; in Bayesian language, these are the likelihood and the prior. Pourya, El Rawas, and Unser introduce FLOWER, a pretrained flow-matching solver for linear inverse problems [1]. Its three steps combine a predicted clean destination, a measurement-aware refinement, and a time-progression update. The algorithm has a proximal-optimization interpretation and an approximate Bayesian posterior-sampling interpretation, each relying on specific assumptions. An important practical issue is that the noise level supplied to the solver may be inaccurate. We therefore ask how reconstruction quality changes when $\sigma_a \neq \sigma_{\text{true}}$, using controlled experiments on several forward operators and the metrics PSNR, SSIM, and LPIPS. The published 100-image FLOWER benchmark serves as context, whereas within-image comparisons with the correctly specified run serve as our experimental control.

2 Background and Related Work

2.1 Bayesian linear inverse problems

Let $x \in \mathbb{R}^d$ denote the unknown clean image and $y \in \mathbb{R}^m$ the observed measurement. The linear Gaussian model is

$$y = Hx + n, \qquad H : \mathbb{R}^d \to \mathbb{R}^m, \qquad n \sim \mathcal{N}(0, \sigma_{\text{true}}^2 I_m). \tag{1}$$

The likelihood and posterior satisfy

$$p(y \mid x) \propto \exp\left(-\frac{\|Hx - y\|_2^2}{2\sigma_{\text{true}}^2}\right), \qquad p(x \mid y) \propto p(y \mid x)p(x). \tag{2}$$

The maximum-a-posteriori estimator minimizes

$$x_{\text{MAP}} \in \arg\min_x \left\{ \frac{\|Hx - y\|_2^2}{2\sigma_{\text{true}}^2} - \log p(x) \right\}. \tag{3}$$

The posterior mean minimizes expected squared error, whereas posterior sampling retains the diversity of the conditional distribution. A higher PSNR is therefore not sufficient evidence of more accurate posterior sampling.

2.2 Flow matching and differential dynamics

Let p_{X_0} be an easy-to-sample source distribution and p_{X_1} the target image distribution. Flow matching transports p_{X_0} toward p_{X_1} through the ordinary differential equation

$$\frac{d\psi_t(x)}{dt} = v_t(\psi_t(x)), \qquad t \in [0,1], \qquad \psi_0(x) = x. \tag{4}$$

The ideal flow-matching loss is

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{X_t\sim p_{X_t}}\left[\|v_t^\theta(X_t) - v_t(X_t)\|_2^2\right]. \tag{5}$$

Because the marginal velocity v_t is generally unavailable, conditional flow matching uses paired endpoints $(X_0, X_1) \sim \pi$ and straight paths:

$$X_t = (1-t)X_0 + tX_1, \qquad \frac{dX_t}{dt} = X_1 - X_0. \tag{6}$$

The practical conditional objective is

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t,(X_0,X_1)\sim\pi}\left[\left\|v_t^\theta((1-t)X_0 + tX_1, t) - (X_1 - X_0)\right\|_2^2\right]. \tag{7}$$

The coupling $\pi \in \Pi(p_{X_0}, p_{X_1})$ specifies how source and target samples are paired. Independent coupling means

$$\pi_{\text{IND}} = p_{X_0} \otimes p_{X_1}. \tag{8}$$

Optimal transport instead seeks

$$\pi^\star \in \arg\min_{\pi\in\Pi(p_{X_0},p_{X_1})} \mathbb{E}_{(X_0,X_1)\sim\pi}\left[\|X_1 - X_0\|_2^2\right]. \tag{9}$$

In practice, mini-batch OT approximates this coupling. The distinction matters: the Bayesian derivation of FLOWER assumes independent source and target variables, whereas its main image benchmarks primarily use mini-batch-OT checkpoints. Unconditional flow sampling may be discretized using the Euler update

$$x_{t+\Delta t} = x_t + \Delta t\, v_t^\theta(x_t). \tag{10}$$

FLOWER changes this unconditional trajectory rather than directly applying Equation (10) to the inverse problem.

2.3 Proximal refinement and related solvers

For a suitable convex function f, the proximal operator is

$$\operatorname{prox}_{\lambda f}(z) = \arg\min_x \left\{ f(x) + \frac{1}{2\lambda}\|x - z\|_2^2 \right\}. \tag{11}$$

The quadratic term preserves proximity to z, while f enforces a desired property. Plug-and-play (PnP) methods combine measurement-consistency steps with learned image priors. PnP-Flow applies this idea to flow matching [3]. FLOWER shares the prior/refinement alternation but motivates its iteration through approximate conditional sampling. Related methods include DPS, ΠGDM, DDS, DiffPIR, DAPS, OT-ODE, D-Flow, and Flow-Priors. Our experiments study the noise-scale sensitivity of FLOWER itself rather than re-ranking these solvers.

3 The FLOWER Method

3.1 Step 1: Estimate the clean destination

At time t, FLOWER predicts a clean destination at $t = 1$:

$$\widehat{x}_1(x_t) = x_t + (1 - t)v_t^\theta(x_t). \tag{12}$$

This is an endpoint prediction, not a direct update to $x_{t+\Delta t}$.

Proposition 1 (Conditional destination)**.** *If v_t^θ attains the population optimum of conditional flow matching, then*

$$\widehat{x}_1(x_t) = \mathbb{E}[X_1 \mid X_t = x_t]. \tag{13}$$

Proof. The minimizer of squared-error conditional regression is

$$v_t^\star(x_t) = \mathbb{E}[X_1 - X_0 \mid X_t = x_t]. \tag{14}$$

Equation (6) implies

$$X_1 = X_t + (1 - t)(X_1 - X_0). \tag{15}$$

Taking conditional expectations,

$$\begin{aligned} \mathbb{E}[X_1 \mid X_t = x_t] &= \mathbb{E}[X_t + (1 - t)(X_1 - X_0) \mid X_t = x_t] && (16)\\ &= x_t + (1 - t)\mathbb{E}[X_1 - X_0 \mid X_t = x_t] && (17)\\ &= x_t + (1 - t)v_t^\star(x_t), && (18) \end{aligned}$$

which equals Equation (12) when $v_t^\theta = v_t^\star$. □

The identity is exact at the optimal velocity field; a finite pretrained network approximates it.

3.2 Step 2: Refine the destination using the measurement

FLOWER approximates the unknown endpoint conditional by

$$\widetilde{p}(x_1 \mid x_t) = \mathcal{N}\left(\widehat{x}_1(x_t), \nu_t^2 I\right), \qquad \nu_t = \frac{1 - t}{\sqrt{t^2 + (1 - t)^2}}. \tag{19}$$

The refinement assumes the likelihood

$$p(y \mid x_1) = \mathcal{N}(y; Hx_1, \sigma_a^2 I), \tag{20}$$

where σ_a denotes the standard deviation supplied to the solver. Bayes' rule gives

$$\widetilde{p}(x_1 \mid x_t, y) \propto p(y \mid x_1)\widetilde{p}(x_1 \mid x_t). \tag{21}$$

Proposition 2 (Gaussian measurement refinement)**.** *Under the preceding Gaussian approximation and linear measurement model,*

$$\widetilde{p}(x_1 \mid x_t, y) = \mathcal{N}(x_1; \mu_t, \Sigma_t), \tag{22}$$

where

$$\Sigma_t = \left(\nu_t^{-2} I + \sigma_a^{-2} H^\top H\right)^{-1}, \tag{23}$$

$$\mu_t = \Sigma_t \left(\nu_t^{-2} \widehat{x}_1 + \sigma_a^{-2} H^\top y\right). \tag{24}$$

Proof. Conditional independence of Y and X_t given X_1 and Bayes' rule yield

$$\widetilde{p}(x_1 \mid x_t, y) = \frac{p(y \mid x_1)\widetilde{p}(x_1 \mid x_t)}{\widetilde{p}(y \mid x_t)}. \tag{25}$$

Taking logarithms and discarding constants independent of x_1,

$$\begin{aligned} -2\log \widetilde{p}(x_1 \mid x_t, y) &= \sigma_a^{-2}(y - Hx_1)^\top (y - Hx_1) \\ &\quad + \nu_t^{-2}(x_1 - \widehat{x}_1)^\top (x_1 - \widehat{x}_1) + C \qquad (26) \\ &= x_1^\top A_t x_1 - 2x_1^\top b_t + C', \qquad (27) \end{aligned}$$

where

$$A_t = \nu_t^{-2} I + \sigma_a^{-2} H^\top H, \tag{28}$$

and

$$b_t = \nu_t^{-2} \widehat{x}_1 + \sigma_a^{-2} H^\top y. \tag{29}$$

Since A_t is positive definite, completing the square gives

$$\begin{aligned} x_1^\top A_t x_1 - 2x_1^\top b_t &= (x_1 - A_t^{-1} b_t)^\top A_t (x_1 - A_t^{-1} b_t) \\ &\quad - b_t^\top A_t^{-1} b_t. \end{aligned} \tag{30}$$

Thus $\widetilde{p}(x_1 \mid x_t, y)$ is Gaussian with covariance

$$A_t^{-1} = \Sigma_t, \tag{31}$$

and mean

$$A_t^{-1} b_t = \mu_t. \tag{32}$$

□

The mean is also the unique minimizer of

$$Q_t(x) = \frac{\|Hx - y\|_2^2}{2\sigma_a^2} + \frac{\|x - \widehat{x}_1\|_2^2}{2\nu_t^2}. \tag{33}$$

So the result is equivalently a proximal refinement:

$$\mu_t = \mathrm{prox}_{\nu_t^2 F_y}(\widehat{x}_1), \qquad F_y(x) = \frac{\|Hx - y\|_2^2}{2\sigma_a^2}. \tag{34}$$

In the general algorithm the refined endpoint is

$$\widetilde{x}_1 = \mu_t + \gamma \kappa_t, \qquad \kappa_t \sim \mathcal{N}(0, \Sigma_t), \qquad \gamma \in \{0, 1\}. \tag{35}$$

The paper draws κ_t without an explicit covariance square root:

$$\kappa_t = \Sigma_t \left(\nu_t^{-1} \epsilon_1 + \sigma_a^{-1} H^\top \epsilon_2 \right), \tag{36}$$

where

$$\epsilon_1 \sim \mathcal{N}(0, I_d), \qquad \epsilon_2 \sim \mathcal{N}(0, I_m). \tag{37}$$

The image-restoration configuration in our experiments uses $\gamma = 0$, so $\widetilde{x}_1 = \mu_t$. Fresh source noise is still introduced in Step 3.

3.3 Step 3: Advance along the flow path

With $\Delta t = 1/N$, the update is

$$x_{t+\Delta t} = (1 - t - \Delta t)\epsilon + (t + \Delta t)\widetilde{x}_1(x_t, y), \qquad \epsilon \sim p_{X_0} = \mathcal{N}(0, I_d). \tag{38}$$

A fresh source sample is drawn at each iteration; its coefficient decreases as time approaches one.

Proposition 3 (Transition induced by refreshed source noise)**.** *Assume a standard Gaussian source and an approximate endpoint law*

$$\widetilde{p}(x_1 \mid x_t, y) = \mathcal{N}(x_1; \mu_t, \Sigma_t). \tag{39}$$

If the next state uses an independent fresh source draw as in Equation (38), its conditional sampling kernel is

$$\widetilde{q}_t(x_{t+\Delta t} \mid x_t, y) = \mathcal{N}\left(x_{t+\Delta t}; (t + \Delta t)\mu_t, (t + \Delta t)^2 \Sigma_t + (1 - t - \Delta t)^2 I_d \right). \tag{40}$$

Gaussian convolution. Conditional on the newly drawn endpoint $\widetilde{X}_1 = x_1$, the fresh source noise is independent and the next state has law $\mathcal{N}((t + \Delta t)x_1, (1 - t - \Delta t)^2 I_d)$. Hence

$$\begin{aligned} \widetilde{q}_t(x_{t+\Delta t} \mid x_t, y) &= \int_{\mathbb{R}^d} \widetilde{p}(x_1 \mid x_t, y) \widetilde{q}_t(x_{t+\Delta t} \mid x_1, x_t, y)\, dx_1 && (41) \\ &= \int_{\mathbb{R}^d} \mathcal{N}(x_1; \mu_t, \Sigma_t) \\ &\quad \times \mathcal{N}\left(x_{t+\Delta t}; (t + \Delta t)x_1, (1 - t - \Delta t)^2 I_d \right) dx_1. && (42) \end{aligned}$$

Writing the second Gaussian in residual form gives

$$\begin{aligned} \widetilde{q}_t(x_{t+\Delta t} \mid x_t, y) = \int_{\mathbb{R}^d} & \mathcal{N}(x_1; \mu_t, \Sigma_t) \\ & \times \mathcal{N}\left(x_{t+\Delta t} - (t + \Delta t)x_1; 0, (1 - t - \Delta t)^2 I_d \right) dx_1. \end{aligned} \tag{43}$$

Set

$$z = (t + \Delta t)x_1. \tag{44}$$

Applying the linear change-of-variables rule to the first Gaussian turns the integral into

$$\begin{aligned} \widetilde{q}_t(x_{t+\Delta t} \mid x_t, y) = \int_{\mathbb{R}^d} & \mathcal{N}\left(z; (t + \Delta t)\mu_t, (t + \Delta t)^2 \Sigma_t \right) \\ & \times \mathcal{N}\left(x_{t+\Delta t} - z; 0, (1 - t - \Delta t)^2 I_d \right) dz. \end{aligned} \tag{45}$$

The Gaussian convolution identity is

$$\int_{\mathbb{R}^d} \mathcal{N}(z;\mu_1,\Sigma_1)\mathcal{N}(x-z;\mu_2,\Sigma_2)\,dz = \mathcal{N}(x;\mu_1+\mu_2,\Sigma_1+\Sigma_2). \tag{46}$$

Using it in Equation (45) gives

$$\widetilde{q}_t(x_{t+\Delta t} \mid x_t, y) = \mathcal{N}\left(x_{t+\Delta t}; (t+\Delta t)\mu_t, (t+\Delta t)^2\Sigma_t + (1-t-\Delta t)^2 I_d\right), \tag{47}$$

which is the desired expression.

□

Equation (38) is the sampling update corresponding to this Gaussian-convolution construction. The paper denotes this kernel by an approximate transition $\widetilde{p}$. The displayed Gaussian law is exact for the *refreshed-noise construction*, but it is not generally the exact conditional transition of one straight-line pair (X_0, X_1): at $t < 1$, conditioning on (X_t, X_1) fixes $X_0 = (X_t - tX_1)/(1-t)$. Thus independence of X_0 and X_1 does not make X_0 independent of X_t after conditioning on X_1.

3.4 Ancestral-sampling interpretation

Theorem 1 (Conditional marginal propagation)**.** *If*

$$X_t \sim p_{X_t|Y=y} \tag{48}$$

and the next state is drawn from the correct transition

$$p_{X_{t+\Delta t}|X_t=x_t,Y=y}, \tag{49}$$

then

$$X_{t+\Delta t} \sim p_{X_{t+\Delta t}|Y=y}. \tag{50}$$

Proof. By marginalization and the chain rule for conditional joint distributions,

$$p(x_{t+\Delta t} \mid y) = \int_{\mathbb{R}^d} p(x_t, x_{t+\Delta t} \mid y)\,dx_t \tag{51}$$

$$= \int_{\mathbb{R}^d} p(x_t \mid y)p(x_{t+\Delta t} \mid x_t, y)\,dx_t. \tag{52}$$

Sampling from these two factors therefore produces the required next conditional marginal.

□

Iteration transports conditional marginals to $t = 1$, provided the procedure starts with

$$X_0 \sim p_{X_0|Y=y}. \tag{53}$$

Under independent source–target coupling,

$$p(X_0 \mid y) = p(X_0), \tag{54}$$

so initialization from Gaussian source noise is permitted.

The theorem above uses the *true* conditional transition. FLOWER instead uses the approximate refreshed-noise kernel, so the theorem alone does not prove exact posterior sampling for the

implemented update. Its Bayesian motivation further invokes an optimal velocity field, the linear Gaussian likelihood, the endpoint Gaussian approximation, and $\gamma = 1$.
The released OT checkpoints and $\gamma = 0$ setting used for image restoration are not exact posterior sampling under those idealized assumptions.

4 Research Question and Experimental Design

4.1 Noise-scale mismatch and its local effect

Our experiment fixes the realized measurement and varies only the noise standard deviation supplied to FLOWER.
Define

$$r = \frac{\sigma_a}{\sigma_{\text{true}}}, \qquad \sigma_a^{-2} = r^{-2}\sigma_{\text{true}}^{-2}. \tag{55}$$

Thus, the assumed likelihood precision is r^{-2} times the true precision.

Table 1: Noise-scale ratios, corresponding variances and likelihood precisions.

r	0.50	0.75	1.00	1.25	1.50
Variance ratio r^2	0.25	0.5625	1.00	1.5625	2.25
Relative likelihood precision r^{-2}	4.00	1.78	1.00	0.64	0.44

For example, $r = 0.5$ assumes half the correct standard deviation, one quarter of its variance, and four times the correct precision.
Evenly spaced standard-deviation ratios are not symmetric in likelihood precision.
For denoising, $H = I$, so Equation (24) reduces to

$$\mu_t = w_t y + (1 - w_t)\widehat{x}_1, \qquad w_t = \frac{\nu_t^2}{\nu_t^2 + \sigma_a^2}, \qquad \frac{\partial w_t}{\partial \sigma_a} = -\frac{2\sigma_a \nu_t^2}{(\nu_t^2 + \sigma_a^2)^2} < 0. \tag{56}$$

For a fixed destination estimate, underestimation therefore increases the measurement weight, while overestimation increases reliance on the flow prediction.
This local identity does not imply monotonic final PSNR because each correction also changes future trajectory states and destinations.

4.2 Controlled paired experiment

We ask whether a wrong noise scale causes a meaningful reconstruction loss, whether effects differ by forward operator, and whether all three metrics agree.
Each experiment uses the true corruption

$$y = Hx + n, \qquad n \sim \mathcal{N}(0, \sigma_{\text{true}}^2 I), \tag{57}$$

and the solver receives

$$\sigma_a = r\sigma_{\text{true}}, \qquad r \in \{0.50, 0.75, 1.00, 1.25, 1.50\}. \tag{58}$$

We select 20 images without replacement from each dataset's 100-image paper-benchmark pool (selection seed 2026).
The same images are used for all tasks and ratios within each dataset.

CelebA faces are center-cropped to 178 pixels and resized to 128×128; AFHQ-Cat images have resolution 256×256.

The selected filenames are stored in the repository manifests.

For each image–task pair, one measurement is generated, and all five ratios share this measurement, the degradation operator or mask, and the source-noise draws used by the solver.

In total,

$$2 \text{ datasets} \times 20 \text{ images} \times 5 \text{ tasks} \times 5 \text{ ratios} = 1000 \text{ reconstructions}. \tag{59}$$

Table 2: Experimental settings. Noise scales and degradations follow the original FLOWER image benchmark. The iteration counts N_C, N_A are for CelebA and AFHQ-Cat, respectively.

Task	σ_{true}	CelebA operator	AFHQ-Cat operator	N_C	N_A
Denoising	0.20	Identity	Identity	100	100
Gaussian deblurring	0.05	61×61 kernel, $\sigma_b = 1$	61×61 kernel, $\sigma_b = 3$	100	500
Super-resolution	0.05	$2\times$ subsampling	$4\times$ subsampling	100	500
Random inpainting	0.01	70% missing	70% missing	100	200
Box inpainting	0.05	Central 40×40 box	Central 80×80 box	100	100

Noise standard deviations refer to the model's $[-1, 1]$ image coordinates.

Mapping image values to $[0, 1]$ halves an additive noise standard deviation before clipping.

The implementation solves the quadratic normal equation for Step 2 using pointwise formulas for identity/masks, an FFT-domain formula for circular Gaussian blur, and a pointwise formula on the sampling support for super-resolution.

Released OT checkpoints and paper-aligned step counts are used throughout.

4.3 Metrics and uncertainty

PSNR measures pixel fidelity (higher is better); SSIM measures structural similarity (higher is better); and LPIPS compares learned perceptual features (lower is better).

LPIPS follows the public-code convention used in the completed experiments: postprocessed images are mapped to $[-1, 1]$ and evaluated with `normalize=True`. This differs from the usual LPIPS input normalization, but the convention is identical across all tested ratios. Appendix B documents the evaluation implementation.

For any metric M, the paired image-level and averaged differences are

$$\Delta M_i(r) = M_i(r) - M_i(1), \qquad \overline{\Delta M}(r) = \frac{1}{20} \sum_{i=1}^{20} \Delta M_i(r). \tag{60}$$

The graphs reproduce the stored approximate 95% normal intervals

$$\overline{\Delta M}(r) \ \pm \ 1.96 \frac{s_{\Delta M}(r)}{\sqrt{20}}. \tag{61}$$

They describe variability across the selected images in this paired run.

They are not simultaneous confidence bands, are not adjusted for multiple comparisons, and do not measure uncertainty over independent solver seeds.

At $r = 1$, the paired difference is exactly zero.

5 Numerical Results

5.1 Correct-noise control and the published benchmark

Table 3 compares our $r = 1$ means over 20 selected images with the published Flower1-OT PSNR means over 100 images.

Differences are useful as a reproducibility check but cannot be attributed to mismatch.

All robustness conclusions use paired within-image changes from $r = 1$.

Table 3: Correct-noise control ($r = 1$) compared with published Flower1-OT benchmark. The last column is an absolute benchmark gap, not a mismatch effect.

Dataset	Task	PSNR	SSIM	LPIPS	Paper PSNR	Gap (dB)
CelebA	Denoising	32.22	.917	.031	32.28	-0.06
	Deblurring	34.74	.947	.025	34.98	-0.24
	Super-resolution	32.14	.923	.033	32.36	-0.22
	Random inpainting	32.65	.943	.017	33.08	-0.43
	Box inpainting	30.54	.946	.021	31.19	-0.65
AFHQ-Cat	Denoising	31.31	.875	.109	31.69	-0.38
	Deblurring	27.45	.751	.297	28.64	-1.19
	Super-resolution	26.15	.732	.287	26.23	-0.08
	Random inpainting	32.15	.912	.046	32.97	-0.82
	Box inpainting	26.70	.917	.061	26.19	+0.51

5.2 PSNR response across the five tasks

Figure 1 shows strong task dependence.

At $r = 0.5$, denoising loses 2.10 dB on CelebA and 1.25 dB on AFHQ-Cat.

In contrast, super-resolution gains 0.42 dB and 0.47 dB, respectively.

Random inpainting is nearly flat, with losses of only 0.05–0.06 dB under the strongest overestimation.

At both $r = 1.25$ and $r = 1.5$, overestimation lowers mean PSNR in every dataset–task combination.

To summarize sensitivity without cancellations between positive and negative aggregate changes, define

$$S_{d,k} = \frac{1}{4} \sum_{r \neq 1} \left| \overline{\Delta \mathrm{PSNR}}_{d,k}(r) \right|. \tag{62}$$

CelebA and AFHQ-Cat denoising have the largest sensitivity values (1.05 dB and 0.93 dB), whereas random inpainting has the smallest (0.02 dB and 0.03 dB).

This is the mean absolute change of the aggregate curve, not the mean of individual-image absolute changes.

5.3 Structural and perceptual agreement

Figures 3 and 4 show that the three metrics do not always agree.

Super-resolution at $r = 0.5$ improves PSNR and SSIM and reduces LPIPS in both datasets.

AFHQ-Cat denoising at $r = 0.75$ also improves all three.

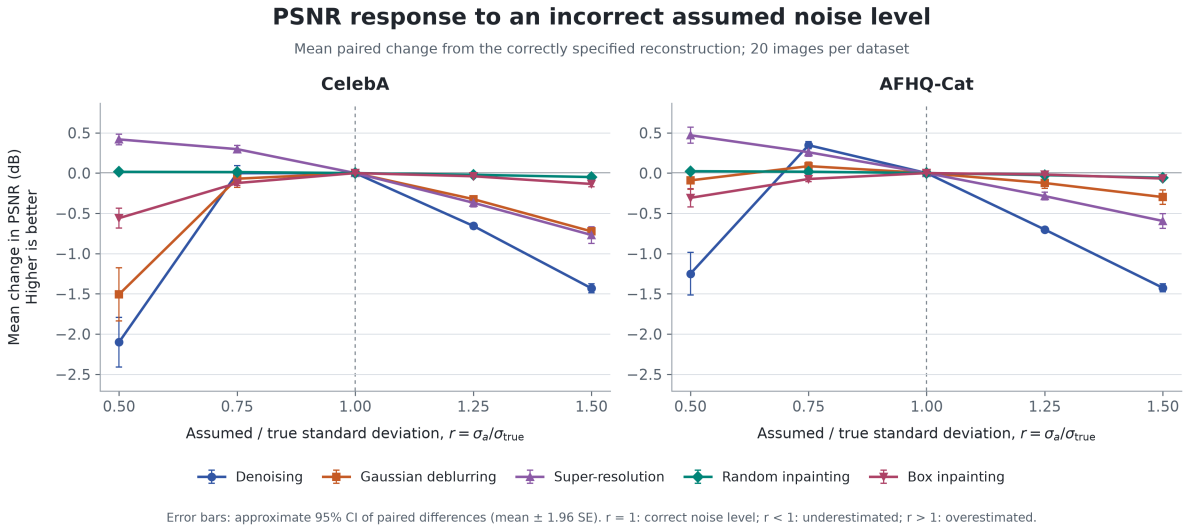


Figure 1: Paired PSNR changes for CelebA and AFHQ-Cat (20 images each). Error bars are approximate 95% intervals across images.

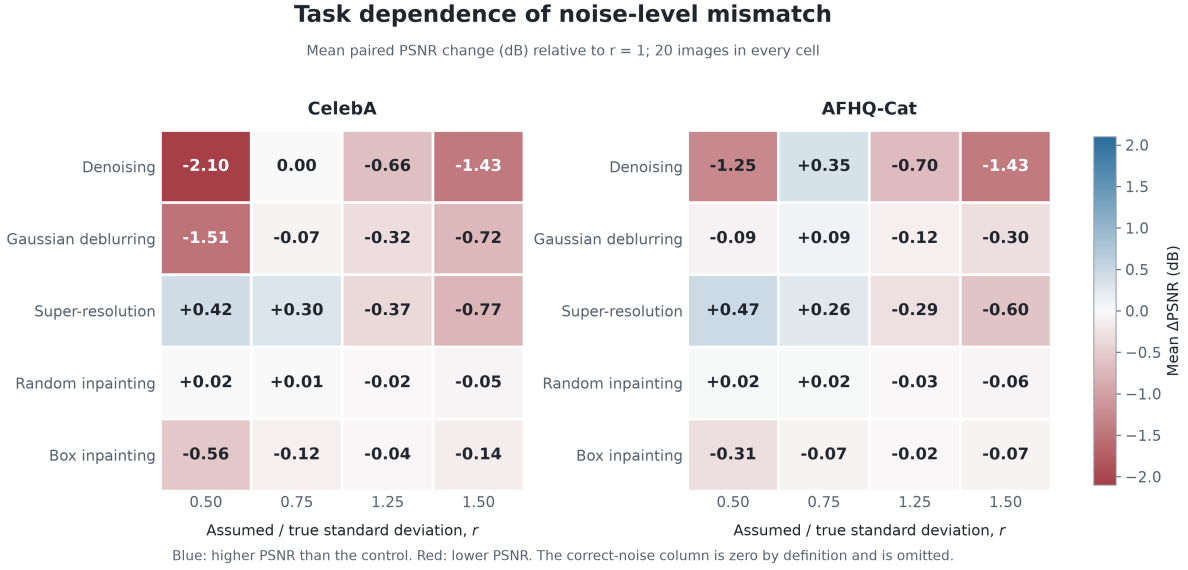


Figure 2: Mean PSNR differences by inverse problem and assumed noise ratio. Positive differences indicate improved fidelity compared with the correct-noise control.

At $r = 0.5$, however, AFHQ-Cat denoising has worse PSNR and SSIM but better LPIPS, illustrating a perceptual–distortion trade-off.

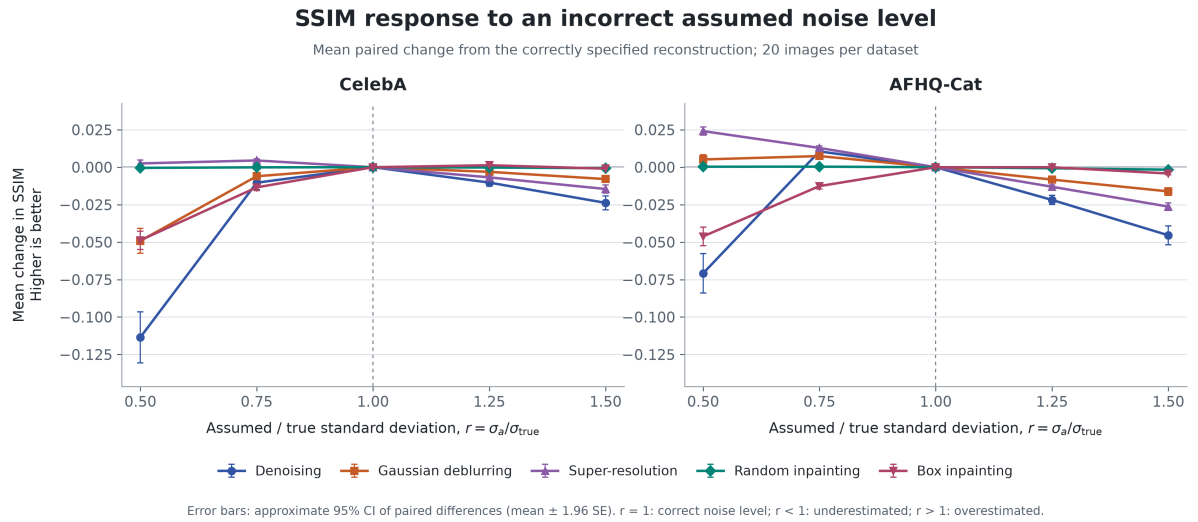


Figure 3: Paired SSIM differences. Higher values indicate improved structural similarity.

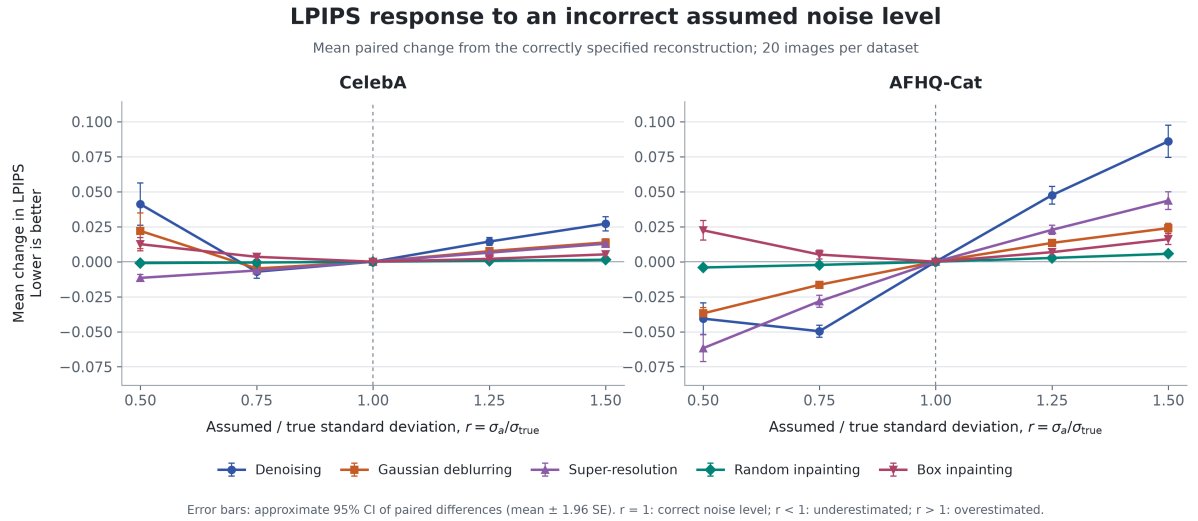


Figure 4: Paired LPIPS differences. Lower values indicate improved perceptual similarity.

5.4 Qualitative reconstruction comparisons

Figures 5 and 6 present fixed images under all five noise assumptions and tasks.

Blue frames mark $r = 1$; each task uses the same measurement and source-noise draws across ratios.

The displayed *measurement image* is $H^\top y$, back-projected into image coordinates for visualization, whereas FLOWER actually receives y.

The examples illustrate the aggregate behavior but do not substitute for 20-image statistics.

Table 4: Selected paired metric changes. PSNR and SSIM improve upward; LPIPS improves downward.

Dataset	Task	r	ΔPSNR (dB)	ΔSSIM	ΔLPIPS
CelebA	Denoising	.50	-2.101	-.1135	+.0411
AFHQ-Cat	Denoising	.50	-1.250	-.0708	-.0406
AFHQ-Cat	Denoising	.75	+.347	+.0106	-.0496
CelebA	Super-resolution	.50	+.419	+.0025	-.0115
AFHQ-Cat	Super-resolution	.50	+.470	+.0241	-.0618
CelebA	Random inpainting	1.50	-.049	-.0006	+.0014
AFHQ-Cat	Random inpainting	1.50	-.058	-.0016	+.0057

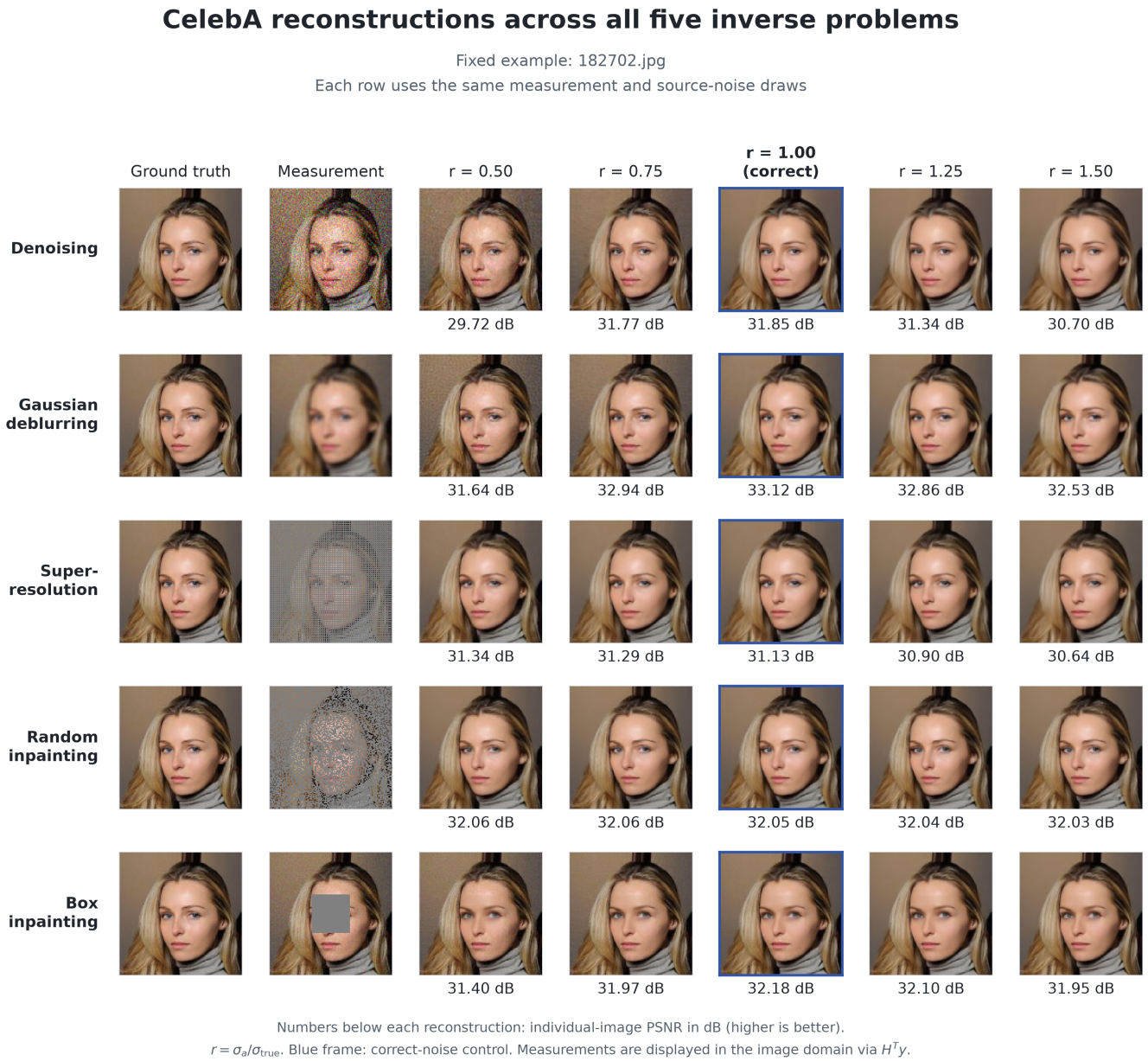


Figure 5: CelebA qualitative comparison, image `182702.jpg`. Values beneath reconstructions are single-image PSNR, not dataset averages.

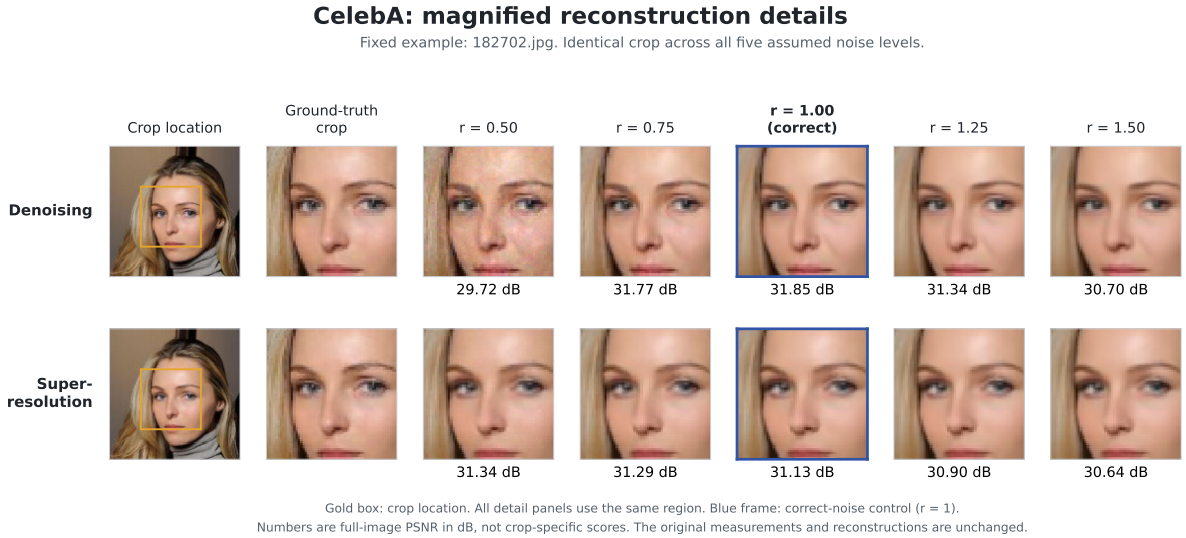


Figure 5: CelebA qualitative comparison (continued): magnified details for denoising and super-resolution from the same image as above. Gold boxes mark the crop location; all ratios use the identical region. The numbers remain full-image PSNR values, not crop-specific scores.



Figure 6: AFHQ-Cat qualitative comparison, image `pixabay_cat_001348.jpg`, with the same column interpretation.

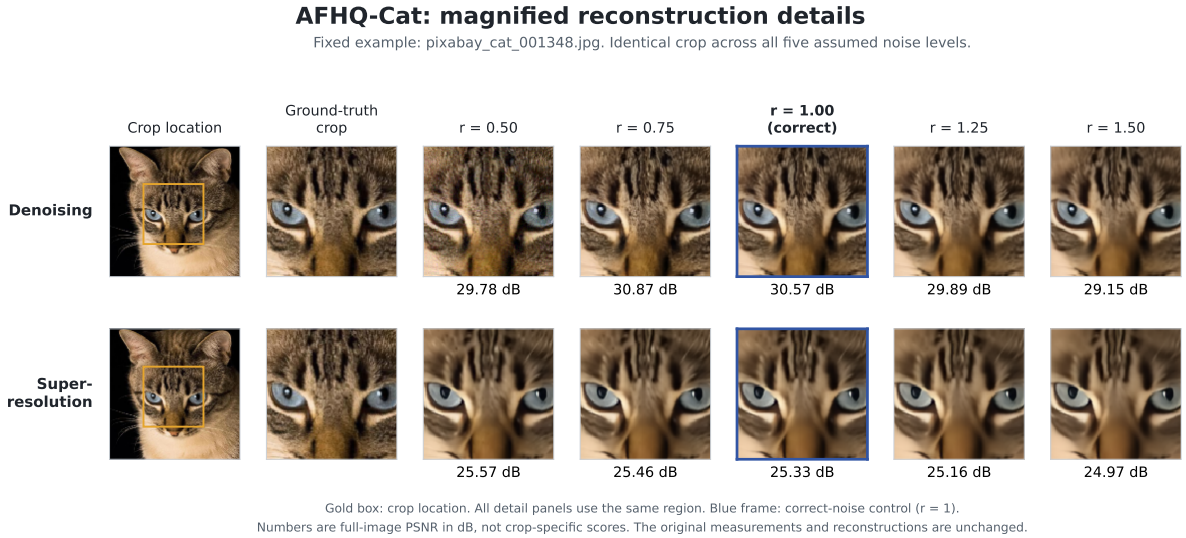


Figure 6: AFHQ-Cat qualitative comparison (continued): magnified details for denoising and super-resolution from the same image as above. Gold boxes mark the crop location; all ratios use the identical region. The numbers remain full-image PSNR values, not crop-specific scores.

5.5 What changes along the reconstruction trajectory

Figure 7 follows a representative CelebA denoising image, `182734.jpg`, selected near the median image-level sensitivity in the most sensitive setting.

Early states contain strong source noise; changes between noise assumptions become clearer as its coefficient shrinks.

Figure 8 isolates the three steps at $t = 0.75$: the noise-scale assumption enters Step 2 directly but also affects later Step 1 estimates indirectly through the changed trajectory.

Step 3 is a noisy state, not a clean endpoint estimate.

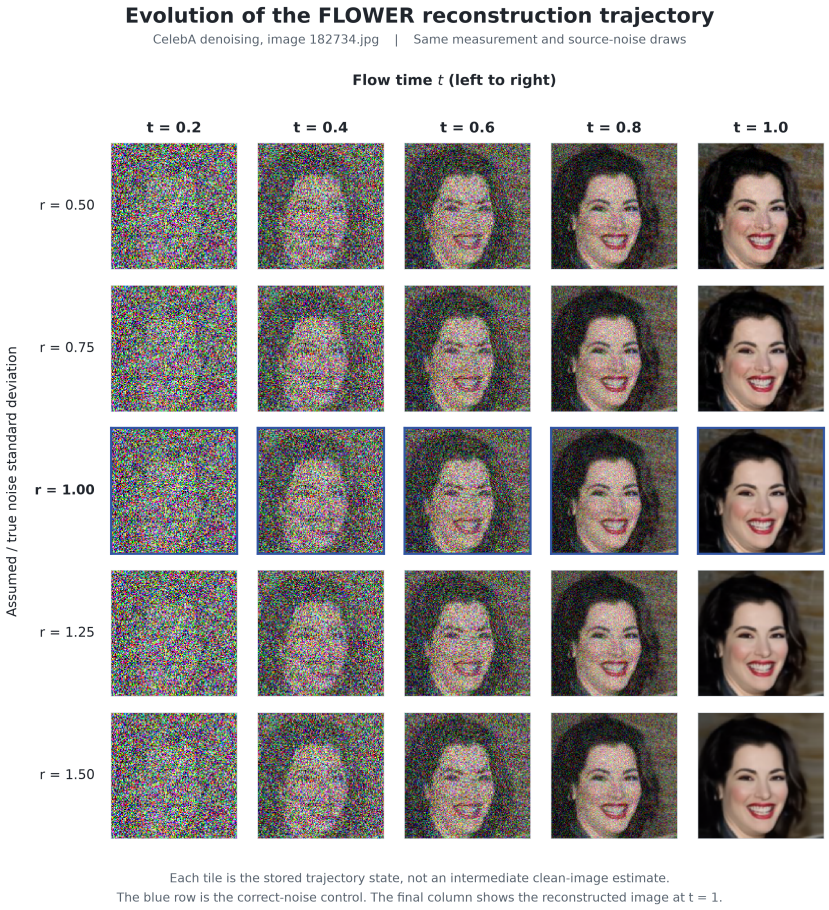


Figure 7: FLOWER trajectory states for CelebA denoising across time and noise-scale ratios; the last column is the final reconstruction.

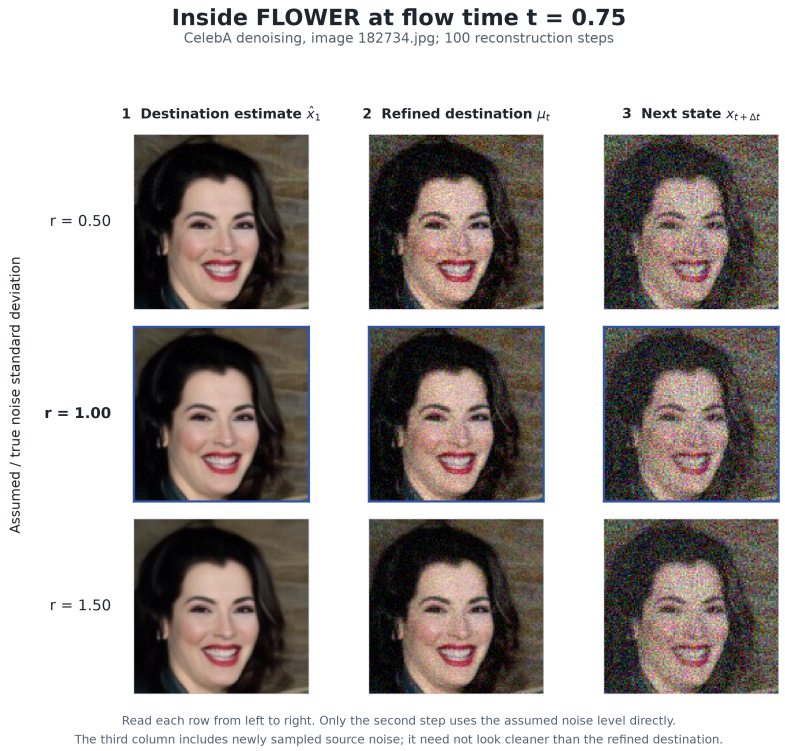


Figure 8: Inside FLOWER at $t = 0.75$: clean destination estimate, measurement-refined destination, and noisy next state.

6 Discussion

6.1 Operator dependence and calibration

The study supports an operator-dependent, rather than universal, conclusion.

Denoising can lose more than 2 dB under severe noise underestimation, while random inpainting is nearly invariant within the tested range.

Equation (56) explains the denoising response: every pixel is directly involved in the measurement term, so a wrong σ_a changes the refinement throughout the image.

For a binary inpainting mask, the data term directly constrains only observed coordinates; unobserved coordinates are primarily supplied by the learned destination.

Effects can still propagate to later iterations.

Random inpainting also uses the smallest true noise scale, $\sigma_{\text{true}} = 0.01$.

Its apparent robustness cannot therefore be attributed to the operator alone: task identity and true noise level co-vary.

The super-resolution gains under underestimation may compensate for practical approximations in the reconstruction method, but the present study does not isolate whether the cause lies in the learned prior, the Gaussian endpoint approximation, discretization, or some other implementation detail.

6.2 Scope and limitations of the evidence

Only 20 fixed images per dataset and one paired random sequence per image–task pair are used.

The reported intervals therefore reflect selected image variability, not independent solver-seed variability or guarantees across datasets.

We tested moderate positive scalar noise ratios rather than unknown noise families or misspecified spatial covariance.
The ratio that maximizes a metric on this sample is descriptive, not a validated noise-calibration rule.
Differences from published 100-image benchmark averages cannot be conflated with within-image mismatch effects.

6.3 Potential extension to nonlinear inverse problems

The original FLOWER paper sketches a nonlinear measurement model $y = h(x_1) + n$ with Gaussian noise.
The Gaussian endpoint approximation then produces

$$\widetilde{p}(x_1 \mid x_t, y) \propto \exp\left[-\frac{\|y - h(x_1)\|_2^2}{2\sigma_a^2} - \frac{\|x_1 - \widehat{x}_1\|_2^2}{2\nu_t^2}\right], \tag{63}$$

which is generally non-Gaussian.
Its score is available:

$$\nabla_{x_1} \log \widetilde{p}(x_1 \mid x_t, y) = \sigma_a^{-2} J_h(x_1)^\top (y - h(x_1)) - \nu_t^{-2}(x_1 - \widehat{x}_1), \tag{64}$$

where J_h denotes the Jacobian of h.
The paper suggests replacing the closed-form Step 2 by an iterative sampler, such as Langevin dynamics.
Our linear experiments do not test robustness of this nonlinear extension.

7 Conclusion

FLOWER combines a pretrained flow-based destination estimate, a Gaussian measurement refinement, and a source-noise time-progression step.
The refinement exposes a precise mathematical mechanism by which noise-scale misspecification changes reconstructions.
Its posterior-sampling motivation should be distinguished from the practical OT and $\gamma = 0$ restoration configuration.
In our paired study, severe underestimation harms denoising, whereas super-resolution improves under underestimation on both datasets.
Random inpainting is nearly insensitive over the tested ratios, and overestimation reduces average PSNR in every tested dataset–task combination.
PSNR, SSIM, and LPIPS sometimes disagree; consequently, no single metric fully characterizes robustness.
Correct physical noise calibration need not maximize every image-quality metric of an approximate practical solver.

A Additional Mathematical Details

A.1 The Gaussian variance schedule

Suppose

$$X_0, X_1 \sim \mathcal{N}(0, I) \tag{65}$$

are independent and

$$X_t = (1-t)X_0 + tX_1. \tag{66}$$

Then

$$\mathrm{Cov}(X_t) = [t^2 + (1-t)^2]I, \qquad \mathrm{Cov}(X_1, X_t) = tI. \tag{67}$$

Gaussian conditioning gives

$$\begin{aligned} \mathrm{Cov}(X_1 \mid X_t) &= I - tI([t^2 + (1-t)^2]I)^{-1}tI && (68)\\ &= \left(1 - \frac{t^2}{t^2 + (1-t)^2}\right) I && (69)\\ &= \frac{(1-t)^2}{t^2 + (1-t)^2} I = \nu_t^2 I. && (70) \end{aligned}$$

This exact result for Gaussian endpoints motivates the schedule, but does not establish isotropic endpoint uncertainty for natural images.

A.2 Sampling the refinement perturbation

Let

$$\epsilon_1 \sim \mathcal{N}(0, I_d), \qquad \epsilon_2 \sim \mathcal{N}(0, I_m) \tag{71}$$

be independent.
Define

$$\kappa_t = \Sigma_t \left(\nu_t^{-1}\epsilon_1 + \sigma_a^{-1} H^\top \epsilon_2\right). \tag{72}$$

It is Gaussian with zero mean and covariance

$$\begin{aligned} \mathrm{Cov}(\kappa_t) &= \Sigma_t \,\mathrm{Cov}(\nu_t^{-1}\epsilon_1 + \sigma_a^{-1} H^\top \epsilon_2)\Sigma_t^\top && (73)\\ &= \Sigma_t(\nu_t^{-2} I + \sigma_a^{-2} H^\top H)\Sigma_t^\top && (74)\\ &= \Sigma_t \Sigma_t^{-1} \Sigma_t = \Sigma_t. && (75) \end{aligned}$$

Thus

$$\kappa_t \sim \mathcal{N}(0, \Sigma_t) \tag{76}$$

without explicitly computing $\Sigma_t^{1/2}$.
When $\gamma = 0$, this perturbation is omitted.

A.3 Known non-isotropic Gaussian noise

For general known covariance $R_n \succ 0$,

$$n \sim \mathcal{N}(0, R_n), \tag{77}$$

the data-fidelity term is

$$F_y(x) = \frac{1}{2}(y - Hx)^\top R_n^{-1}(y - Hx). \tag{78}$$

The Gaussian posterior-refinement formulas become

$$\Sigma_t = (\nu_t^{-2} I + H^\top R_n^{-1} H)^{-1}, \tag{79}$$

$$\mu_t = \Sigma_t(\nu_t^{-2}\widehat{x}_1 + H^\top R_n^{-1} y). \tag{80}$$

This is the covariance-aware extension described in the original paper.

Our experiment does not vary the covariance structure R_n independently of the scalar noise scale.

B Evaluation and Reproducibility Details

Numerical results reside in `artifacts/experiments/main/`.

`metrics.csv` contains the 1,000 image-level rows; `metrics_with_deltas.csv` contains paired differences; and `summary.csv` contains the 50 dataset–task–ratio aggregates.

`run_metadata.json` stores the experimental settings and seeds, while selected filenames are stored under `data_splits/`.

Selected qualitative and trajectory PNGs are provided, but not a PNG for every numerical reconstruction.

PSNR and SSIM are computed on clipped $[0, 1]$ images.

PSNR uses `skimage` with data range 1 and SSIM uses the PyTorch-Ignite implementation.

LPIPS uses AlexNet and reproduces the public-code convention: postprocessed images are mapped to $[-1, 1]$ and the call sets `normalize=True`.

This differs from the usual LPIPS convention, but it is identical across all tested ratios.

The stored CSV metrics are the numerical source of truth; the displayed PNGs underwent quantization.

The uncertainty intervals reproduce the original stored normal-approximation summaries.

C Complete Numerical Results

Each entry in Tables 5 and 6 is averaged over 20 fixed images.

The $r = 1$ rows define the correctly specified control; ΔPSNR is paired with the same image's control.

Table 5: Complete CelebA results (20 fixed images per row).

Task	r	PSNR	SSIM	LPIPS	ΔPSNR
Denoising	.50	30.12	.803	.072	-2.101
	.75	32.22	.906	.024	-0.001
	1.00	32.22	.917	.031	0.000
	1.25	31.57	.906	.045	-0.657
	1.50	30.79	.893	.058	-1.432
Deblurring	.50	33.24	.898	.047	-1.507
	.75	34.67	.941	.020	-0.070
	1.00	34.74	.947	.025	0.000

Task	r	PSNR	SSIM	LPIPS	ΔPSNR
	1.25	34.42	.944	.032	-0.324
	1.50	34.02	.939	.039	-0.724
Super-resolution	.50	32.55	.926	.021	+0.419
	.75	32.43	.928	.027	+0.297
	1.00	32.14	.923	.033	0.000
	1.25	31.77	.917	.039	-0.369
	1.50	31.37	.909	.046	-0.769
Random inpainting	.50	32.66	.943	.016	+0.016
	.75	32.66	.943	.016	+0.012
	1.00	32.65	.943	.017	0.000
	1.25	32.63	.943	.018	-0.021
	1.50	32.60	.942	.018	-0.049
Box inpainting	.50	29.98	.897	.033	-0.560
	.75	30.41	.932	.024	-0.124
	1.00	30.54	.946	.021	0.000
	1.25	30.50	.947	.023	-0.040
	1.50	30.40	.945	.026	-0.135

Table 6: Complete AFHQ-Cat results (20 fixed images per row).

Task	r	PSNR	SSIM	LPIPS	ΔPSNR
Denoising	.50	30.06	.805	.068	-1.250
	.75	31.65	.886	.059	+0.347
	1.00	31.31	.875	.109	0.000
	1.25	30.61	.854	.157	-0.702
	1.50	29.88	.830	.195	-1.426
Deblurring	.50	27.36	.756	.260	-0.091
	.75	27.54	.759	.280	+0.086
	1.00	27.45	.751	.297	0.000
	1.25	27.33	.743	.310	-0.123
	1.50	27.15	.735	.321	-0.298
Super-resolution	.50	26.62	.756	.225	+0.470
	.75	26.41	.745	.259	+0.259
	1.00	26.15	.732	.287	0.000
	1.25	25.86	.719	.310	-0.286
	1.50	25.56	.706	.331	-0.595
Random inpainting	.50	32.17	.912	.042	+0.023
	.75	32.17	.912	.043	+0.017
	1.00	32.15	.912	.046	0.000
	1.25	32.12	.911	.048	-0.026
	1.50	32.09	.910	.051	-0.058
Box inpainting	.50	26.39	.871	.084	-0.307
	.75	26.62	.905	.066	-0.074
	1.00	26.70	.917	.061	0.000
	1.25	26.68	.917	.068	-0.018

Task	r	PSNR	SSIM	LPIPS	ΔPSNR
	1.50	26.63	.913	.077	-0.068

D Earlier Internal States

Figures 9 and 10 repeat the three-step diagnostic at earlier flow times, using the same source-noise and measurement controls as in Figure 8.



Figure 9: Three internal FLOWER steps at $t = 0.25$.

E Metric Comparisons by Inverse Problem

These panels compare datasets for one inverse problem at a time.

Each metric uses its own vertical scale, making small differences easier to distinguish than in the shared overview plots.

The numerical values and confidence intervals are the same as those in Section 5.

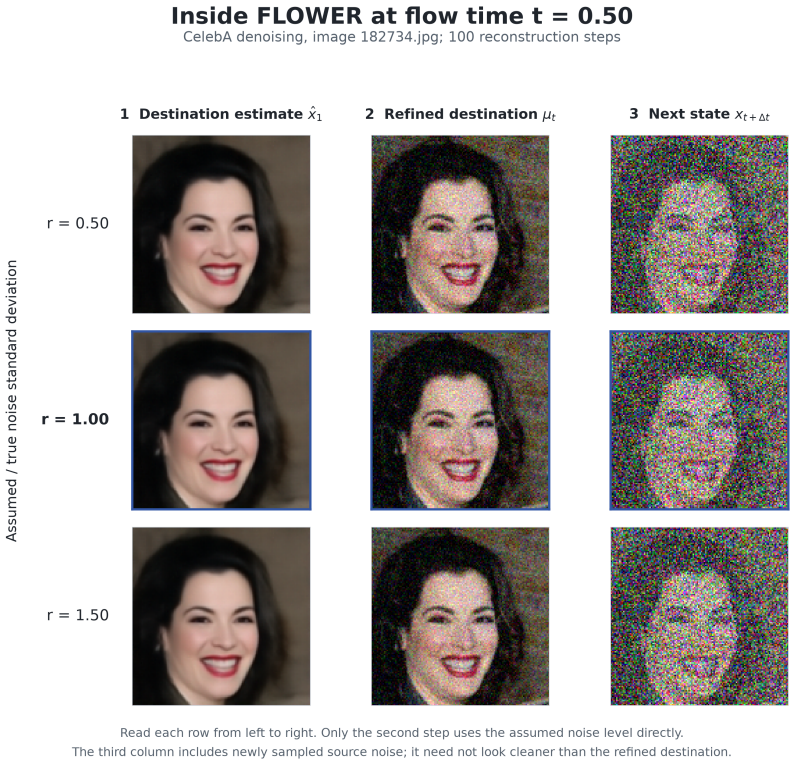


Figure 10: Three internal FLOWER steps at $t = 0.50$.

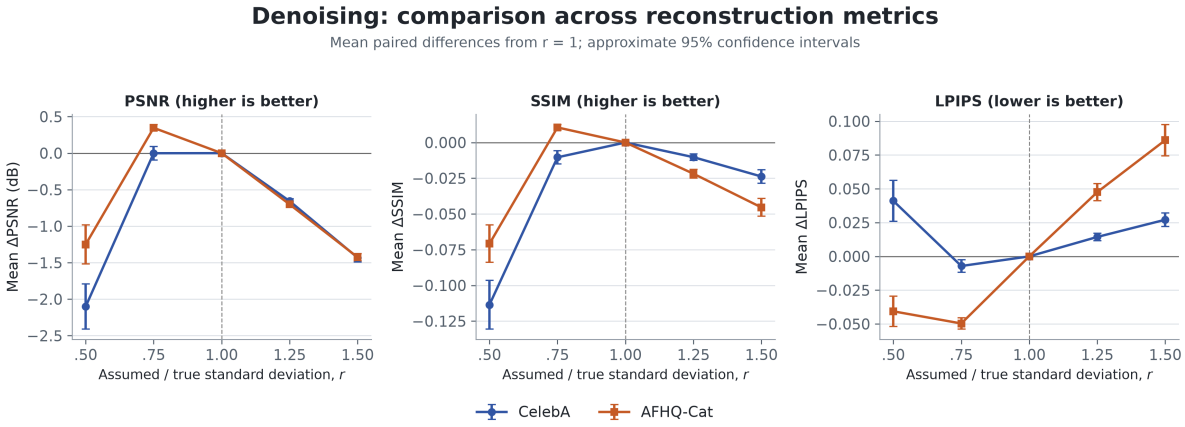


Figure 11: Denoising across PSNR, SSIM, and LPIPS. AFHQ-Cat shows a perceptual–distortion trade-off at $r = 0.5$.

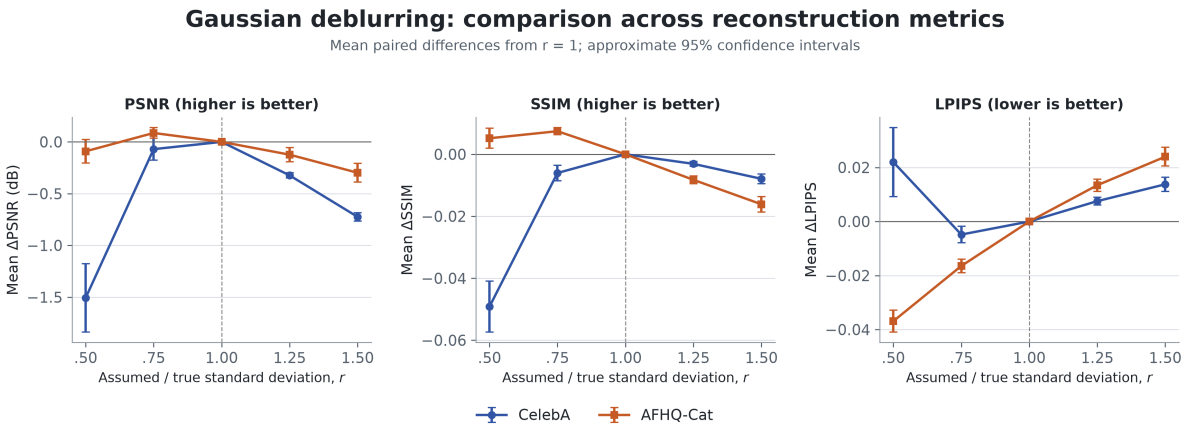


Figure 12: Gaussian deblurring across PSNR, SSIM, and LPIPS.

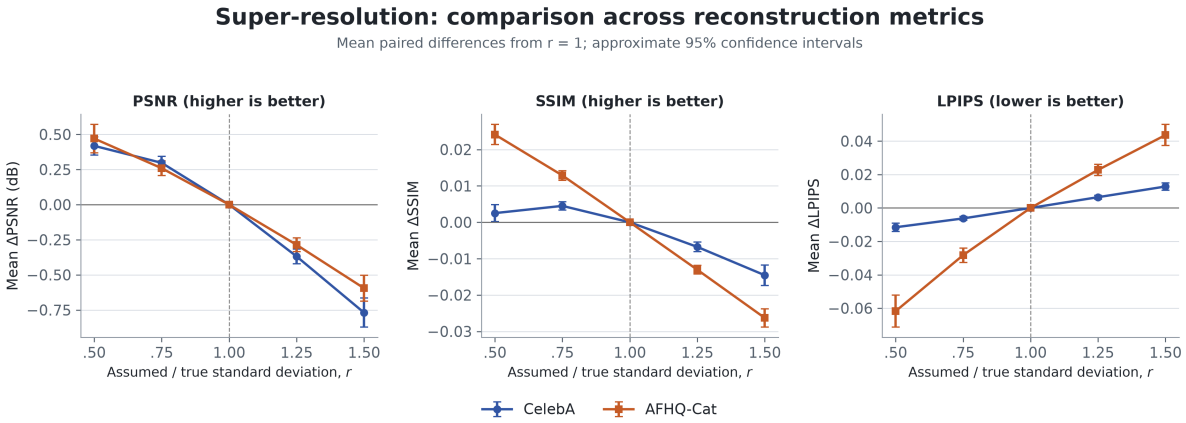


Figure 13: Super-resolution across PSNR, SSIM, and LPIPS. All three metrics improve at $r = 0.5$ on both datasets.

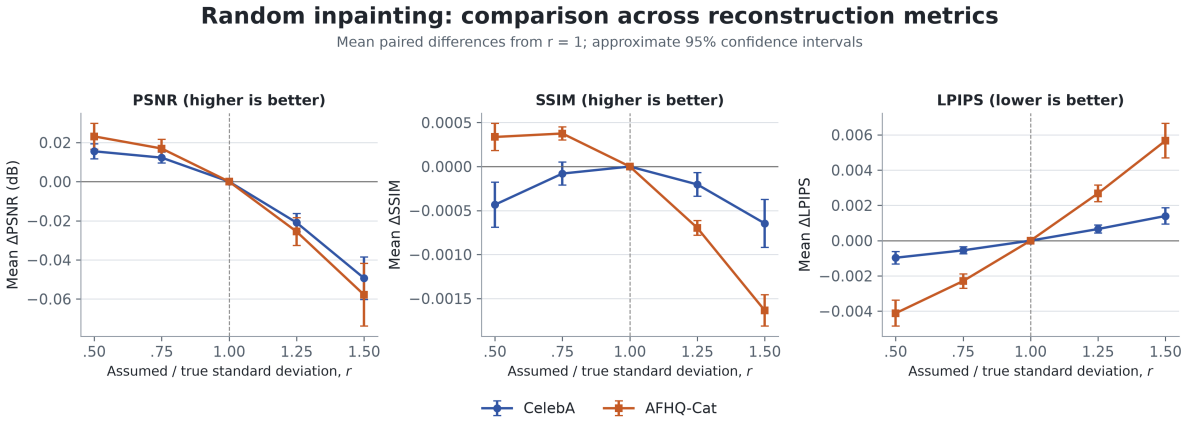


Figure 14: Random inpainting across PSNR, SSIM, and LPIPS. Changes are small in magnitude.

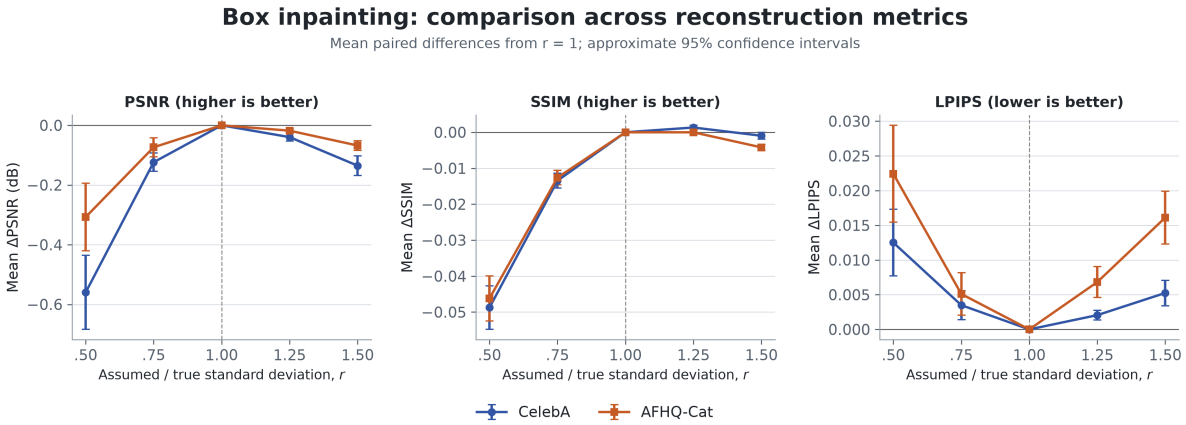


Figure 15: Box inpainting across PSNR, SSIM, and LPIPS. Underestimation can affect SSIM and LPIPS even where PSNR changes are modest.

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